Department of Computer Science And Engineering/AD/DS/AI/CI

ASSIGNMENT- II
Subject Name:**Design and Analysis of Algorithm**

Course Code: BCS401

Date of Assignment31/05/2026

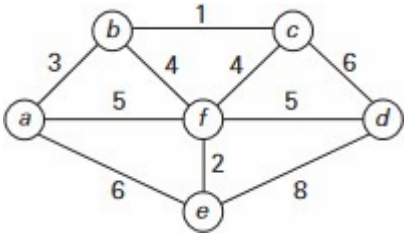
Date of submission: 10/06/2026

Faculty Name:BARNALI CHAKRABORTY **Max. Marks: 25**

SEC: 'A''B''C'

Module3 and Module4 Module5

1. Define AVL Trees. Explain its four rotation types 10M
2. Construct bottom up heap for the list 2,9,7,6,5,8. Obtain its time complexity 10
3. Define heap. Explain the properties of heap along with its representation. 10
   Design Horspools algorithm for string matching. Apply Horspools algorithm to find
4. the pattern BARBER in the text: JIM_SAW_ME_IN_A_BARBERSHOP 10
   5.Construct minimum cost spanning tree using Kruskals algorithm for the following graph.

5. 

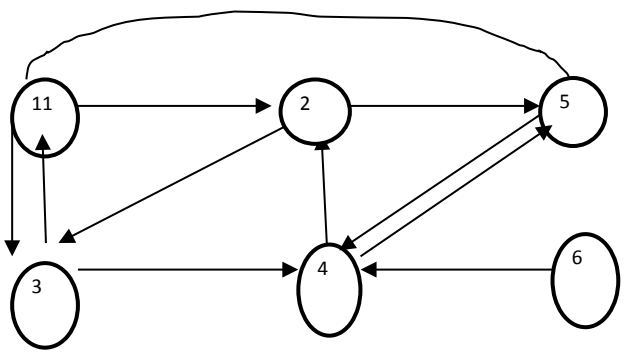
10

6. What are Huffman Trees? Construct the Huffman tree for the following data.

Character	A	B	C	D	E	-
Probability	0.5	0.35	0.5	0.1	0.4	0.2

Encode DAD-CBE using Huffman Encoding. 10

7.Apply Dijkstra's algorithm to find single source shortest path for the given graph by considering S as the source vertex.



0	15	10	999	45	999
999	0	15	999	20	999
20	999	0	20	999	0
999	10	999	0	35	999
999	999	999	30	0	999
999	999	999	4	999	0

Intial Matrix 10

20 4

8.Define transitive closure of a graph. Apply Warshalls algorithm to compute transitive closure of a directed graph

	a	*b*	*c*	*d*
a	0	1	0	0
b	0	0	0	1
c	0	0	0	0
d	1	0	1	0

10

9.Explain the following with examples

i) P problem
ii) NP Problem
iii) NP- Complete problem

NP – Hard Problems

10

10 , What is backtracking? Apply backtracking to solve the below instance of sum of subset problem S={5,10,12,13,15,18} d=30 10

11. Illustrate N queen's problem using backtracking to solve 4-Queens problem 10
12. Using Branch and Bound technique solve the below instance of knapsack problem.

Item	Weight	Value
1	4	40
2	7	42
3	5	25
4	3	12

Capacity 5

13. Explain Space and trade offs. Give the algorithm for sorting by Comparison Counting. Sort the given arrays using sorting comparison. 25 45 10 20 50 15. 10
14.Explain Branch and Bound . Knapsack Approximation Algorithm. 10
15.Explain Decision Tree with an example, 10

Write any one between 16 0r 17 and 18 0r 19

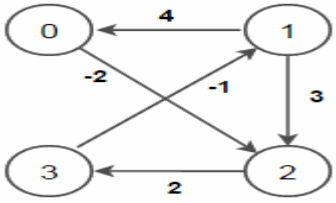
16.Write an Floyd's algorithm and find time complexity .
Solve the all pairs shortest path problem for the given adjacency matrix as shown below.

	a	b	c	d
a	0	infinity	3	infinity
b	2	0	infinity	infinity
c	infinity	7	0	1
d	6	infinity	infinity	0

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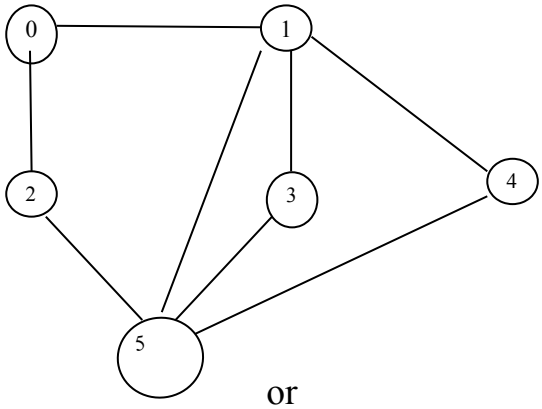
or

17.Explain Floyd's algorithm with pseudo code and find all-pair shortest path for the given digraph



Write any one

18.Prim's problem:



$$\begin{pmatrix} 0 & 60 & 10 & \infty & \infty & \infty \\ 60 & 0 & \infty & 20 & 40 & 70 \\ 10 & \infty & 0 & \infty & \infty & 50 \\ \infty & 20 & \infty & 0 & \infty & 80 \\ \infty & 40 & \infty & \infty & 0 & 30 \\ \infty & 70 & 50 & 80 & 30 & 0 \end{pmatrix}$$

or

19.Write prims algorithm and give its time complexity.

Note:

1. ***Submission*** *of two* ***assignments for each course*** *is mandatory.*
2. *The grade or marks that you get on your* ***assignments will*** *be counted in your final CIA.*
3. *Late submission of the assignment may lead to a deduction of 20% of the* ***marks.***